Problem 9.11. Consider a particle of charge q and mass m, free to move in the xy plane in response to an electromagnetic wave propagating in the z direction (Eq. 9.49 – might as well set $\delta = 0$).

(a) Ignoring the magnetic force, find the velocity of the particle, as a function of time. (Assume the average velocity is zero.)
(b) Now calculate the resulting magnetic force on the particle.
(c) Show that the (time) average magnetic force is *zero*.

The problem with this naïve model for the pressure of light is that the velocity is $90°$ out of phase with the fields. For energy to be absorbed, there's got to be some *resistance* to the motion of the charges. Suppose we include a force of the form $-\gamma m\mathbf{v}$, for some damping constant γ.

(d) Repeat part (a) (ignore the exponentially damped transient). Repeat part (b), and find the average magnetic force on the particle.[12]

Problem 9.12. In the complex notation there is a clever device for finding the time average of a product. Suppose $f(\mathbf{r}, t) = A\cos(\mathbf{k}\cdot\mathbf{r} - \omega t + \delta_a)$ and $g(\mathbf{r}, t) = B\cos(\mathbf{k}\cdot\mathbf{r} - \omega t + \delta_b)$. Show that $\langle fg\rangle = (1/2)\,\mathrm{Re}(\tilde{f}\tilde{g}^*)$, where the star denotes complex conjugation. [Note that this only works if the two waves have the same $\mathbf{k}$ and ω, but they need not have the same amplitude or phase.] For example,

$$\langle u\rangle = \frac{1}{4}\,\mathrm{Re}\left(\epsilon_0\tilde{\mathbf{E}}\cdot\tilde{\mathbf{E}}^* + \frac{1}{\mu_0}\tilde{\mathbf{B}}\cdot\tilde{\mathbf{B}}^*\right) \quad \text{and} \quad \langle\mathbf{S}\rangle = \frac{1}{2\mu_0}\,\mathrm{Re}(\tilde{\mathbf{E}}\times\tilde{\mathbf{B}}^*).$$

Problem 9.13. Find all elements of the Maxwell stress tensor for a monochromatic plane wave traveling in the z direction and linearly polarized in the x direction (Eq. 9.49). Does your answer make sense? (Remember that $-\overleftrightarrow{\mathbf{T}}$ represents the momentum flux density.) How is the momentum flux density related to the energy density, in this case?

9.3 Electromagnetic Waves in Matter

9.3.1 Propagation in Linear Media

Inside matter, but in regions where there is no *free* charge or *free* current, Maxwell's equations become

$$\left.\begin{aligned} &\text{(i)}\ \ \boldsymbol{\nabla}\cdot\mathbf{D} = 0, \quad \text{(iii)}\ \ \boldsymbol{\nabla}\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t}, \\ &\text{(ii)}\ \ \boldsymbol{\nabla}\cdot\mathbf{B} = 0, \quad \text{(iv)}\ \ \boldsymbol{\nabla}\times\mathbf{H} = \frac{\partial\mathbf{D}}{\partial t}. \end{aligned}\right\} \tag{9.66}$$

If the medium is *linear*,

$$\mathbf{D} = \epsilon\mathbf{E}, \quad \mathbf{H} = \frac{1}{\mu}\mathbf{B}, \tag{9.67}$$

[12] C. E. Mungan, *Am. J. Phys.* **77**, 965 (2009). See also Prob. 9.36.

and *homogeneous* (so ϵ and μ do not vary from point to point), they reduce to

$$\left.\begin{aligned} &\text{(i)}\ \ \boldsymbol{\nabla}\cdot\mathbf{E}=0, \quad \text{(iii)}\ \ \boldsymbol{\nabla}\times\mathbf{E}=-\frac{\partial\mathbf{B}}{\partial t},\\ &\text{(ii)}\ \ \boldsymbol{\nabla}\cdot\mathbf{B}=0, \quad \text{(iv)}\ \ \boldsymbol{\nabla}\times\mathbf{B}=\mu\epsilon\frac{\partial\mathbf{E}}{\partial t}, \end{aligned}\right\} \tag{9.68}$$

which differ from the vacuum analogs (Eqs. 9.40) only in the replacement of $\mu_0\epsilon_0$ by $\mu\epsilon$.[13] Evidently, electromagnetic waves propagate through a linear homogeneous medium at a speed

$$v=\frac{1}{\sqrt{\epsilon\mu}}=\frac{c}{n}, \tag{9.69}$$

where

$$n\equiv\sqrt{\frac{\epsilon\mu}{\epsilon_0\mu_0}} \tag{9.70}$$

is the **index of refraction** of the substance. For most materials, μ is very close to μ_0, so

$$n\approx\sqrt{\epsilon_r}, \tag{9.71}$$

where ϵ_r is the dielectric constant[14] (Eq. 4.34). Since ϵ_r is almost always greater than 1, light travels *more slowly* through matter – a fact that is well known from optics.

All of our previous results carry over, with the simple transcription $\epsilon_0\to\epsilon$, $\mu_0\to\mu$, and hence $c\to v$. The energy density is[15]

$$u=\frac{1}{2}\left(\epsilon E^2+\frac{1}{\mu}B^2\right), \tag{9.72}$$

and the Poynting vector is

$$\mathbf{S}=\frac{1}{\mu}(\mathbf{E}\times\mathbf{B}). \tag{9.73}$$

For monochromatic plane waves, the frequency and wave number are related by $\omega=kv$ (Eq. 9.11), the amplitude of $\mathbf{B}$ is $1/v$ times the amplitude of $\mathbf{E}$ (Eq. 9.48), and the intensity[16] is

[13] This observation is mathematically pretty trivial, but the physical implications are extraordinary: As the wave passes through, the fields busily polarize and magnetize all the molecules, and the resulting (oscillating) dipoles create their own electric and magnetic fields. These combine with the original fields in such a way as to create a *single* wave with the same frequency but a different speed. This remarkable conspiracy (known in optics as the **Ewald–Oseen extinction theorem**) is responsible for the phenomenon of **transparency**. It is a distinctly *non*trivial consequence of linearity. For further discussion see M. B. James and D. J. Griffiths, *Am. J. Phys.* **60**, 309 (1992); H. Fearn, D. F. V. James, and P. W. Milonni, *Am. J. Phys.* **64**, 986 (1996); M. Mansuripur, *Opt. Photonics News* **9**, 50 (1998).

[14] The dielectric constant is "constant" in the sense of being independent of the *amplitude* of $\mathbf{E}$, but it may well depend on the *frequency*, as we shall see. Thus, for example, if you quote the (static) dielectric constant for water, from Table 4.2, you will conclude that the index of refraction is 8.9, which is *wildly* off, for visible light ($n=1.33$).

[15] See Prob. 8.27; refer to Section 4.4.3 for the precise *meaning* of "energy density," in the context of linear media.

[16] The *momentum* carried by an electromagnetic wave in matter is controversial. See, for example, S. M. Barnett, *Phys. Rev. Lett.* **104**, 070401 (2010).

$$I = \frac{1}{2}\epsilon v E_0^2. \tag{9.74}$$

The interesting question is this: What happens when a wave passes from one transparent medium into another – air to water, say, or glass to plastic? As in the case of waves on a string, we expect to get a reflected wave and a transmitted wave. The details depend on the exact nature of the electrodynamic boundary conditions, which we derived in Chapter 7 (Eq. 7.74):

$$\left.\begin{aligned} &\text{(i)}\ \ \epsilon_1 E_1^{\perp} = \epsilon_2 E_2^{\perp}, \quad \text{(iii)}\ \ \mathbf{E}_1^{\parallel} = \mathbf{E}_2^{\parallel}, \\ &\text{(ii)}\ \ B_1^{\perp} = B_2^{\perp}, \qquad\ \ \text{(iv)}\ \ \frac{1}{\mu_1}\mathbf{B}_1^{\parallel} = \frac{1}{\mu_2}\mathbf{B}_2^{\parallel}. \end{aligned}\right\} \tag{9.75}$$

These equations relate the electric and magnetic fields on either side of the interface between two linear media. In the following sections, we use them to deduce the laws governing reflection and refraction of electromagnetic waves.

9.3.2 Reflection and Transmission at Normal Incidence

Suppose the xy plane forms the boundary between two linear media. A plane wave of frequency ω, traveling in the z direction and polarized in the x direction, approaches the interface from the left (Fig. 9.13):

$$\left.\begin{aligned} \tilde{\mathbf{E}}_I(z,t) &= \tilde{E}_{0_I} e^{i(k_1 z - \omega t)}\hat{\mathbf{x}}, \\ \tilde{\mathbf{B}}_I(z,t) &= \frac{1}{v_1}\tilde{E}_{0_I} e^{i(k_1 z - \omega t)}\hat{\mathbf{y}}. \end{aligned}\right\} \tag{9.76}$$

It gives rise to a reflected wave

$$\left.\begin{aligned} \tilde{\mathbf{E}}_R(z,t) &= \tilde{E}_{0_R} e^{i(-k_1 z - \omega t)}\hat{\mathbf{x}}, \\ \tilde{\mathbf{B}}_R(z,t) &= -\frac{1}{v_1}\tilde{E}_{0_R} e^{i(-k_1 z - \omega t)}\hat{\mathbf{y}}, \end{aligned}\right\} \tag{9.77}$$

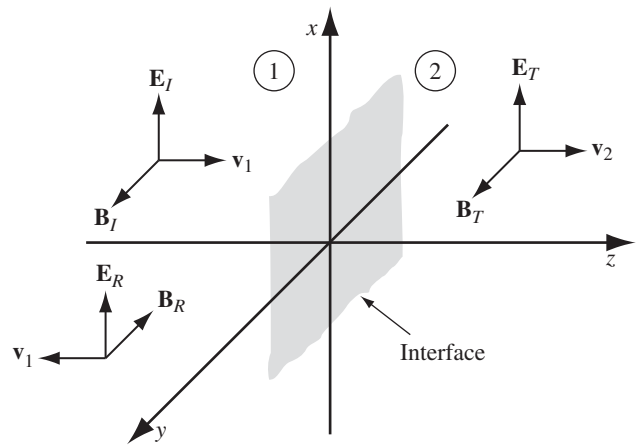


Fig. 9.13

which travels back to the left in medium 1, and a transmitted wave

$$\left.\begin{aligned}\tilde{\mathbf{E}}_T(z,t) &= \tilde{E}_{0_T} e^{i(k_2 z-\omega t)}\hat{\mathbf{x}},\\ \tilde{\mathbf{B}}_T(z,t) &= \frac{1}{v_2}\tilde{E}_{0_T} e^{i(k_2 z-\omega t)}\hat{\mathbf{y}},\end{aligned}\right\} \tag{9.78}$$

which continues on to the right in medium 2. Note the minus sign in $\tilde{\mathbf{B}}_R$, as required by Eq. 9.50 – or, if you prefer, by the fact that the Poynting vector aims in the direction of propagation.

At $z = 0$, the combined fields on the left, $\tilde{\mathbf{E}}_I + \tilde{\mathbf{E}}_R$ and $\tilde{\mathbf{B}}_I + \tilde{\mathbf{B}}_R$, must join the fields on the right, $\tilde{\mathbf{E}}_T$ and $\tilde{\mathbf{B}}_T$, in accordance with the boundary conditions (Eq. 9.75). In this case there are no components perpendicular to the surface, so (i) and (ii) are trivial. However, (iii) requires that

$$\tilde{E}_{0_I} + \tilde{E}_{0_R} = \tilde{E}_{0_T}, \tag{9.79}$$

while (iv) says

$$\frac{1}{\mu_1}\Big(\frac{1}{v_1}\tilde{E}_{0_I} - \frac{1}{v_1}\tilde{E}_{0_R}\Big) = \frac{1}{\mu_2}\Big(\frac{1}{v_2}\tilde{E}_{0_T}\Big), \tag{9.80}$$

or

$$\tilde{E}_{0_I} - \tilde{E}_{0_R} = \beta\tilde{E}_{0_T}, \tag{9.81}$$

where

$$\beta \equiv \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{\mu_1 n_2}{\mu_2 n_1}. \tag{9.82}$$

Equations 9.79 and 9.81 are easily solved for the outgoing amplitudes, in terms of the incident amplitude:

$$\tilde{E}_{0_R} = \Big(\frac{1-\beta}{1+\beta}\Big)\tilde{E}_{0_I}, \quad \tilde{E}_{0_T} = \Big(\frac{2}{1+\beta}\Big)\tilde{E}_{0_I}. \tag{9.83}$$

These results are strikingly similar to the ones for waves on a string. Indeed, if the permeabilities μ are close to their values in vacuum (as, remember, they *are* for most media), then $\beta = v_1/v_2$, and we have

$$\tilde{E}_{0_R} = \Big(\frac{v_2 - v_1}{v_2 + v_1}\Big)\tilde{E}_{0_I}, \quad \tilde{E}_{0_T} = \Big(\frac{2v_2}{v_2 + v_1}\Big)\tilde{E}_{0_I}, \tag{9.84}$$

which are *identical* to Eqs. 9.30. In that case, as before, the reflected wave is *in phase* (right side up) if $v_2 > v_1$ and *out of phase* (upside down) if $v_2 < v_1$; the real amplitudes are related by

$$E_{0_R} = \Big|\frac{v_2 - v_1}{v_2 + v_1}\Big|E_{0_I}, \quad E_{0_T} = \Big(\frac{2v_2}{v_2 + v_1}\Big)E_{0_I}, \tag{9.85}$$

or, in terms of the indices of refraction,

$$E_{0_R} = \Big|\frac{n_1 - n_2}{n_1 + n_2}\Big|E_{0_I}, \quad E_{0_T} = \Big(\frac{2n_1}{n_1 + n_2}\Big)E_{0_I}. \tag{9.86}$$

What fraction of the incident energy is reflected, and what fraction is transmitted? According to Eq. 9.74, the intensity (average power per unit area) is

$$I = \frac{1}{2}\epsilon v E_0^2.$$

If (again) $\mu_1 = \mu_2 = \mu_0$, then the ratio of the reflected intensity to the incident intensity is

$$R \equiv \frac{I_R}{I_I} = \left(\frac{E_{0_R}}{E_{0_I}}\right)^2 = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2, \tag{9.87}$$

whereas the ratio of the transmitted intensity to the incident intensity is

$$T \equiv \frac{I_T}{I_I} = \frac{\epsilon_2 v_2}{\epsilon_1 v_1}\left(\frac{E_{0_T}}{E_{0_I}}\right)^2 = \frac{4n_1 n_2}{(n_1 + n_2)^2}. \tag{9.88}$$

Here, R is the **reflection coefficient** and T the **transmission coefficient**; they tell you the fraction of the incident energy that is reflected and transmitted, respectively. Notice that

$$R + T = 1, \tag{9.89}$$

as conservation of energy, of course, requires. For instance, when light passes from air ($n_1 = 1$) into glass ($n_2 = 1.5$), $R = 0.04$ and $T = 0.96$. No surprise: most of the light is transmitted.

Problem 9.14. Calculate the *exact* reflection and transmission coefficients, *without* assuming $\mu_1 = \mu_2 = \mu_0$. Confirm that $R + T = 1$.

Problem 9.15. In writing Eqs. 9.77 and 9.78, I tacitly assumed that the reflected and transmitted waves have the same *polarization* as the incident wave – along the x direction. Prove that this *must* be so. [*Hint:* Let the polarization vectors of the transmitted and reflected waves be

$$\hat{\mathbf{n}}_T = \cos\theta_T\,\hat{\mathbf{x}} + \sin\theta_T\,\hat{\mathbf{y}}, \quad \hat{\mathbf{n}}_R = \cos\theta_R\,\hat{\mathbf{x}} + \sin\theta_R\,\hat{\mathbf{y}},$$

and prove from the boundary conditions that $\theta_T = \theta_R = 0$.]

9.3.3 Reflection and Transmission at Oblique Incidence

In the previous section, I treated reflection and transmission at *normal* incidence – that is, when the incoming wave hits the interface head-on. We now turn to the more general case of *oblique* incidence, in which the incoming wave meets the boundary at an arbitrary angle θ_I (Fig. 9.14). Of course, normal incidence is really just a special case of oblique incidence, with $\theta_I = 0$, but I wanted to treat it separately, as a kind of warm-up, because the algebra is now going to get a little heavy.

Suppose, then, that a monochromatic plane wave

$$\tilde{\mathbf{E}}_I(\mathbf{r}, t) = \tilde{\mathbf{E}}_{0_I} e^{i(\mathbf{k}_I \cdot \mathbf{r} - \omega t)}, \quad \tilde{\mathbf{B}}_I(\mathbf{r}, t) = \frac{1}{v_1}(\hat{\mathbf{k}}_I \times \tilde{\mathbf{E}}_I) \tag{9.90}$$

approaches from the left, giving rise to a reflected wave,

$$\tilde{\mathbf{E}}_R(\mathbf{r}, t) = \tilde{\mathbf{E}}_{0_R} e^{i(\mathbf{k}_R \cdot \mathbf{r} - \omega t)}, \quad \tilde{\mathbf{B}}_R(\mathbf{r}, t) = \frac{1}{v_1}(\hat{\mathbf{k}}_R \times \tilde{\mathbf{E}}_R), \tag{9.91}$$

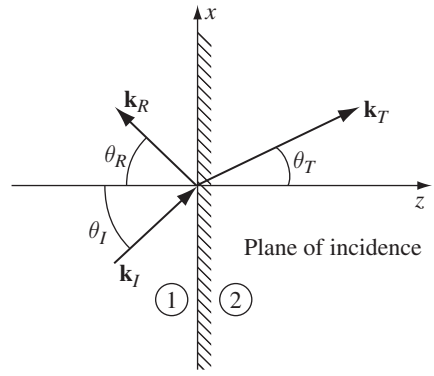


Fig. 9.14

and a transmitted wave

$$\tilde{\mathbf{E}}_T(\mathbf{r},t) = \tilde{\mathbf{E}}_{0_T} e^{i(\mathbf{k}_T\cdot\mathbf{r}-\omega t)}, \quad \tilde{\mathbf{B}}_T(\mathbf{r},t) = \frac{1}{v_2}(\hat{\mathbf{k}}_T \times \tilde{\mathbf{E}}_T). \tag{9.92}$$

All three waves have the same *frequency* ω – which is determined once and for all at the source (the flashlight, or whatever, that produces the incident beam).[17] The three wave numbers are related by Eq. 9.11:

$$k_I v_1 = k_R v_1 = k_T v_2 = \omega, \quad \text{or } k_I = k_R = \frac{v_2}{v_1}k_T = \frac{n_1}{n_2}k_T. \tag{9.93}$$

The combined fields in medium 1, $\tilde{\mathbf{E}}_I + \tilde{\mathbf{E}}_R$ and $\tilde{\mathbf{B}}_I + \tilde{\mathbf{B}}_R$, must now be joined to the fields $\tilde{\mathbf{E}}_T$ and $\tilde{\mathbf{B}}_T$ in medium 2, using the boundary conditions (Eq. 9.75). These all share the generic structure

$$(\ )e^{i(\mathbf{k}_I\cdot\mathbf{r}-\omega t)} + (\ )e^{i(\mathbf{k}_R\cdot\mathbf{r}-\omega t)} = (\ )e^{i(\mathbf{k}_T\cdot\mathbf{r}-\omega t)} \quad \text{at } z = 0. \tag{9.94}$$

I'll fill in the parentheses in a moment; for now, the important thing to notice is that the x, y, and t dependence is confined to the exponents. *Because the boundary conditions must hold at all points on the plane, and for all times, these exponential factors must be equal* (when $z = 0$). Otherwise, a slight change in x, say, would destroy the equality (see Prob. 9.16). Of course, the time factors are *already* equal (in fact, you could regard this as a confirmation that the transmitted and reflected frequencies must match the incident one). As for the spatial terms, evidently

$$\mathbf{k}_I \cdot \mathbf{r} = \mathbf{k}_R \cdot \mathbf{r} = \mathbf{k}_T \cdot \mathbf{r} \quad \text{when } z = 0, \tag{9.95}$$

or, more explicitly,

$$x(k_I)_x + y(k_I)_y = x(k_R)_x + y(k_R)_y = x(k_T)_x + y(k_T)_y \tag{9.96}$$

for all x and all y.

[17] Nonlinear ("active") media can change the frequency, but we are talking only about *linear* media.